EE 446: TinyML — Assignment 2 Written Report

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Problem 3(c): Base Model Evaluation & Metrics

Task Description: Evaluate the unquantized base Deep Neural Network (DNN) model on the test set (`X_test`) using `classification_report` and `confusion_matrix` from the `sklearn.metrics` library.

Confusion Matrix & Classification Report Output

```
788/788 [==============================] - 1s 589us/step
              precision    recall  f1-score   support

           0       1.00      1.00      1.00     11809
           1       1.00      1.00      1.00     13386

    accuracy                           1.00     25195
   macro avg       1.00      1.00      1.00     25195
weighted avg       1.00      1.00      1.00     25195

[[11788    21]
 [   20 13366]]
```

Figure 1: Base model evaluation output screenshot on X test.

Problem 4(b): Full-Integer Quantized Model Evaluation

Task Description: Evaluate the post-training full-integer INT8 quantized TFLite model (`tflite_quant_model`) on the test set (`X_test`) using proper input quantization and scale/zero-point parameters.

Quantized Model Metrics & Impact Analysis

Confusion Matrix & Classification Report Output

```
Classification Report:
              precision    recall  f1-score   support

           0       1.00      1.00      1.00     11809
           1       1.00      1.00      1.00     13386

    accuracy                           1.00     25195
   macro avg       1.00      1.00      1.00     25195
weighted avg       1.00      1.00      1.00     25195

Confusion Matrix:
[[11784    25]
 [   38 13348]]
```

Figure 2: Full-integer quantized model evaluation output screenshot on `X_test`.

Problem 5(c) & 5(d): Arduino Nano 33 BLE Deployment Outputs

Task Description: Hardware deployment and inference execution on the Arduino Nano 33 BLE Sense using TensorFlow Lite for Microcontrollers.

Problem 5(c): Serial Monitor Output for Initial Samples (1–5)

Hardware deployment execution output reporting `Sample #`, `Predicted Class`, and `Actual Class` for the initial batch of 5 test samples.

```
Sample #: 0 | Predicted Class: 1 | Actual Class: 1
Sample #: 1 | Predicted Class: 1 | Actual Class: 1
Sample #: 2 | Predicted Class: 0 | Actual Class: 0
Sample #: 3 | Predicted Class: 0 | Actual Class: 0
Sample #: 4 | Predicted Class: 0 | Actual Class: 1
```

Figure 3: Serial Monitor printouts for samples 1 through 5 on Arduino Nano 33 BLE Sense.

Problem 5(d): Serial Monitor Output for Secondary Test Set (10 Samples)

Deployment verification using the second test feature vector set extracted from `X_test` (excluding the initial 5 samples).

```
Sample #: 0 | Predicted Class: 1 | Actual Class: 1
Sample #: 1 | Predicted Class: 1 | Actual Class: 1
Sample #: 2 | Predicted Class: 1 | Actual Class: 1
Sample #: 3 | Predicted Class: 1 | Actual Class: 1
Sample #: 4 | Predicted Class: 0 | Actual Class: 0
Sample #: 5 | Predicted Class: 0 | Actual Class: 0
Sample #: 6 | Predicted Class: 0 | Actual Class: 0
Sample #: 7 | Predicted Class: 1 | Actual Class: 1
Sample #: 8 | Predicted Class: 1 | Actual Class: 1
Sample #: 9 | Predicted Class: 1 | Actual Class: 1
```

Figure 4: Serial Monitor printouts for 10 additional test samples updated in `network_data.ino`.